# Principle of Inheritance and Variation (DPP)

## DPP - 1

**1.** For Homologues pair of chromosomes which is not true
1) Similar in Morphology
2) Gene for same character present
3) Show pairing in mitosis
4) Show pairing in Meiosis

**2.** Gene is located on –
1) DNA
2) Chromatin
3) Chromosome
4) All of the above

**3.** Tall and dwarf are two traits have common
1) Locus
2) Chromosome
3) Allele
4) gene

**4.** If round seed is one trait, then trait which have common locus is –
1) Plant height
2) Wrinkled seed
3) Green seed
4) yellow seed

**5.** Which of the following traits represent the diploid condition-
1) Axial flower and green seed
2) Round seed and green seed
3) Yellow seed and green seed
4) Axial flower and round seed

**6.** If plant have genotype of AaBbCc than possible genotype of pollen is –
1) AaB, BCЬ
2) ABD, AAB
3) abc, ABD
4) ABc, abC

**7.** Which one from those given below is the period of Mendel's hybridization experiments?
1) 1856-1863
2) 1840-1850
3) 1857-1869
4) 1870-1877

**8.** What contributed to Mendel's success?
i. Selection of pure line pea varieties
ii. Give general principle of inheritance rather unsubstantiated DATA
iii. Selecting one character at a time
iv. Statistical analysis and mathematical logic

Choose the correct option

1) i, ii, iii, and iv
2) ii and iii
3) i, iii and iv
4) ii, iii and iv

**9.** Mark the odd one (with respect to dominant trait in garden pea)
1) Yellow pod
2) Inflated pod
3) Axial flower
4) Yellow seed

**10.** Which law of Mendel explain, why in F2 phenotypic ratio is 3:1
1) Segregation
2) Dominance
3) Independent assortment
4) Polygenic inheritance

**11.** $F_1$ progeny of Mendelian dihybrid cross produces
1) Two types of pollen grains
2) Four genotypes of gametes
3) Two types of eggs and four type of pollen
4) Four types of Diploid pollens

**12.** In monohybrid cross, pure homozygous plants will be.....in F2
1) One
2) Two
3) Three
4) Four

**13.** A dihybrid condition in gamete is
1) TTRRR
2) TtRR
3) Ttrryy
4) TR

**14.** A cross between Aabb X AaBB yields a genotypic ratio of
1) 1 AaBb: 1 aaBb
2) 1 AaBb: 3 aaBb
3) 1 AABb: 2 AaBb: 1 aaBb
4) 1 AABb: 1 aaBb

**15.** Number of phenotype in F1 generation of mendelian dihybrid cross-
1) Two
2) Three
3) One
4) None of the above

## DPP -02

**16.** Hybridization between Tt x tt gives rise to the progeny of ratio-:
1) 1:1
2) 1:2:1
3) 1:2
4) 4:1

**17.** In a given plant, red colour (R) of fruit is dominant over white fruit (r): and tallness (T) is dominant over dwarfness (t). If a plant with genotype RRTt is crossed with a plant of genotype rrtt, what will be the percentage of all tall plants with red fruits in the next generation?
1) 100%
2) 25%
3) 50%
4) 75%

**18.** In order to find out the different types of gametes produced by a pea plant having the genotype AaBb, it should be crossed to a plant with the genotype
1) aaBB
2) AaBb
3) AABB
4) aabb

**19.** In pea plants yellow seeds are dominant to green. If a heterozygous yellow seeded plant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in F1 generation?
1) 50:50
2) 9:1
3) 1:3
4) 3:1

**20.** The offspring of AAbb x aaBB is crossed with aabb. The genotypic ratio of progeny will be
1) 9:3:3:1
2) 1:2:1
3) 1:1:1:1
4) 4:1

**21.** Which is incorrect about mendelian factor's-
1) Occur in pair
2) Unit of inheritance
3) Segregate during gamete formation
4) Show blending

**22.** Which is incorrect about test cross-
1) Type of back cross
2) Give ratio 1:1
3) Give ratio 3:1
4) Cross with recessive parent

**23.** Which result is confirmed by law of dominance –
1) F1 have only one type of progeny
2) F2 have ratio 3:1
3) No blending
4) Both (a) and (b)

**24.** Number of alleles in gamete for two character can be-
1) Three
2) Two
3) Five
4) One

**25.** In 1900 A.D, three biologists independently discovered Mendel's principles. They are
1) De Vries, Correns and Tschermak
2) Sutton, Morgan and Bridges

3) Avery, McLeod and McCarty
4) Bateson, Punnett and Bridges

**26.** Number of pink colour flower obtained by crossing Rr × Rr in a population of 400 in Mirabilis species will be
1) 300
2) 200
3) 100
4) zero

**27.** Identify the wrong statement about Codominance
1) Both alleles belong to same gene loci
2) Both alleles are dominant allele
3) AB blood group is an example
4) Sickle cell anaemia carrier individual is an example

**28.** Select the list of items of Column I with Column II and select the correct option from the codes given below:

| Column I | | Column II | |
|---|---|---|---|
| A | Codominance | i | Starch synthesis gene |
| B | Polygenes | ii | AB Blood group (multiple genes) in humans |
| C | Pleiotropy | iii | Skin colour in humans |

1) A-ii, B-i, C-iii
2) A-ii, B-iii, C-i
3) A-iii, B-ii, C-i
4) A-i, B-iii, C-ii

**29.** Identify the incorrect statement
1) Starch synthesis in pea seed is controlled by one gene
2) Starch grain produced size in seed is example of incomplete dominance
3) Test cross is not type of back cross
4) Dominance is not an autonomous feature of gene

**30.** The modified allele is equivalent to the unmodified allele when it produces-
1) A non-functional enzyme
2) No enzyme
3) The normal enzyme
4) All the above

## DPP-03

**31.** In the example of pleiotropy, the enzyme hydrolase is responsible for the conversion of phenylalanine to
1) Tryptophan
2) Tyrosine
3) Thymine
4) Threonine

**32.** Which of the following pairs of features is a good example of polygenic inheritance?
1) Human height and skin colour
2) ABO blood groups in humans and flower colour of Mirabilis jalapa
3) Hair pigment of mouse and tongue rolling in humans
4) Human eye colour and sickle cell anaemia

**33.** How many gene loci in human controlled skin colour?

1) Two
2) One
3) Three
4) Six

**34.** Which of the following is incorrect regarding pleiotropy-
1) Involve single gene
2) Enzyme whose presence can manifest many phenotype
3) PKU and Sickle cell anaemia is an example
4) PKU is result of many mutations

**35.** S-I: In case of multiple alleles no. of genotype can be possible to predict but not easy to predict no. of phenotype
S-II: Skin colour is Quantitative trait
1) Only S-I correct
2) Only S-II is correct
3) Both S-I and S-II is correct
4) Both wrong

**36.** What will be possible blood group in children from the parents with B and AB blood groups?
1) A, O
2) A, B, AB & O
3) A, B, AB
4) B, O

**37.** Multiple alleles are present:
1) In different chromosomes
2) At different loci on chromosome
3) At the same locus on homologous chromosomes
4) At the non-homologous chromosome

**38.** S-I: In a polygenic trait the phenotype reflects the contribution of each allele.
S-II: The alleles $I^A$ and $I^B$ produces same type of the sugar
1) Only S-I correct
2) Only S-II is correct
3) Both S-I and S-II is correct
4) Both wrong

**39.** Which of the following statements is not true for genes that show 50% recombination frequency?
1) The genes are tightly linked
2) The genes show independent assortment
3) 50 percent parental progeny
4) The genes may be on different chromosomes

**40.** Name of Scientists who proposed that 'pairing and separation' of a pair of chromosomes would lead to separation of factors are-
1) Thomas hunt Morgan
2) Sutton and Boveri
3) De Vries and Correns
4) Punnet and Bateson

**41.** S-I: By 1902, chromosome movement during meiosis had been worked out
S-II: Boveri united the knowledge of chromosomal segregation with mendelian principle and called it the chromosomal theory of inheritance
1) Only S-I is correct
2) Only S-II is correct

3) Both S-I and S-II are correct
4) Both S-I and S-II are wrong

**42.** In case of complete linkage in two gene loci located on same chromosome, result of test cross will give result
1) Recombinant 25 percent
2) Parental 100 percent
3) Recombinant 100 percent
4) Parental 50 percent

**43.** A and B genes are linked. What shall be genotype of progeny in a cross between AaBb and aabb (without crossing over):
1) AAbb and aabb
2) AaBb and aabb
3) AABB and aabb
4) None

**44.** If linked genes are separated through crossing over than probable result is –
1) Parental 100%
2) Recombinant 75%
3) Parental below 50%
4) recombinant below 50%

**45.** If distance between two linked gene A and B is 25 map unit than percentage of parental type is
1) 75 percent
2) 25 percent
3) 50 percent
4) 12 percent

## DPP-04

**46.** Which chromosome set is found in male grasshopper
1) 2A+XY
2) 2A+XO
3) 2A+YY
4) 2A+XX

**47.** ZZ/ZW type of sex determination is seen in
1) Peacock
2) Cockroach
3) Apple snail
4) Human

**48.** Point mutations are caused by
1) Substitution
2) Deletion
3) Insertion
4) All of above

**49.** Mutation can be induced by
1) Radiation
2) Temperature
3) Chemicals
4) All of above

**50.** Example of point mutation is

1) Haemophilia
2) Sickle cell anaemia
3) Thalassemia
4) Colorblindness

**51.** Gene mutation is caused due to
1) Changes in the sequences of genes in DNA
2) Linkage
3) Changes in the sequence of nitrogen bases
4) Reproduction

**52.** Which one of the following is a sexlinked character
1) AB blood group in humans
2) White eye in Drosophila
3) Duffy blood group in human beings
4) Vestigial wing

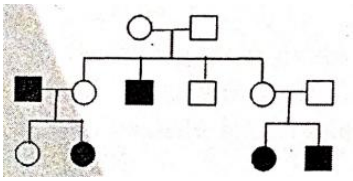
**53.** A genetic disease, which is never passed on from father to son is
1) X-chromosomal linked disease
2) Y-chromosomal linked disease
3) Autosomal linked disease
4) None of above

**54.** When the chromosome humber is different than a multiple of basic chromosome number, it is called
1) Aneuploidy
2) Monoploidy
3) Diploidy
4) Polyploidy

**55.** A trisomic individual has a chromosome number of
1) 2n-1
2) 2n+3
3) 2n +1
4) 2n+2

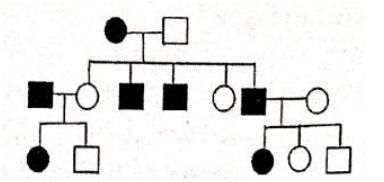
**56.** No. of chromosomes in drone, when male honeybee has 16 chromosomes are:
1) n = 32
2) 2n = 32
3) n = 16
4) 2n = 16

**57.** The genotype of the affected progenies from the given pedigree-



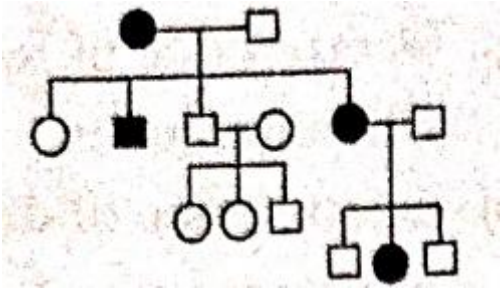
1) Aa
2) Aa
3) AA
4) None of the above

**58.** Which of the following disorder is not possible by studying the given pedigree charts?



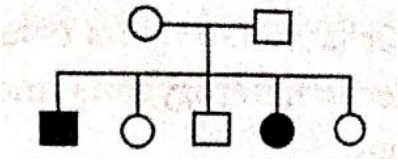
1) X-linked recessive
2) Autosomal recessive
3) Y-linked
4) Autosomal Dominant

**59.** The given pedigree chart is related to which disease:



1) Hypertrichosis
2) Haemophilia
3) Colour blindness
4) Myotonic dystrophy

**60.** Given pedigree chart depicts the inheritance of attached earlobes, an autosomal recessive trait.



Which of the following conclusions drawn is correct?
1) Parents are heterozygous
2) Parents are homozygous dominant.
3) Parents are homozygous recessive.
4) None of these

## DPP-05

**61.** All sons of a couple are colour blind because
1) Mother is homozygous colour blind
2) Mother is heterozygous and father normal
3) Mother is heterozygous and father colour blind
4) Mother is normal and father colour blind

**62.** Colour blindness is a disease usually affecting man but the factor for it is transmitted to the children always by woman. This is because particular factor is located on
1) X chromosome
2) Y chromosome
3) Both X and Y chromosomes
4) Autosome

**63.** If a haemophilic female who marries a normal male gives birth to a male child, then what are the chances of boy being normal.
1) 0%

2) 50%
3) 25%
4) 75%

**64.** A diseased man marries a normal woman, they get three daughters and five sons. All the daughters were diseased and sons were normal. The gene of this disease is
1) Autosomal dominant
2) Sex linked recessive
3) Sex limited character
4) Sex linked dominant

**65.** Klinefelter's syndrome results from the fusion of
1) An X egg and a YY sperm.
2) An XY egg and an X sperm
3) An XX egg and a Y sperm
4) An XX egg and a YY sperm.

**66.** Polyploidy means:
1) Occurrence of haploid set of Chromosomes
2) Occurrence of diploid set of chromosomes
3) Occurrence of more than diploid set of chromosomes
4) None of the above

**67.** In which of the following disorder a single protein that is a part of the cascade of proteins involved in blood clotting is affected?
1) Thalassemia
2) Sickle cell anaemia
3) Haemophilia
4) Phenylketonuria

**68.** Mark the correct statement (w.r.t. stickle cell anemia)
1) Homozygous individuals for HbS are apparently unaffected
2) Heterozygous individuals exhibit sickle cell trait
3) Heterozygous individuals are affected as well as carrier
4) Homozygous individuals for HbA show the diseased phenotype

**69.** In phenylketonuria,
1) Break down of phenylalanine is rapid
2) Accumulation of phenylalanine in body
3) Chromosomal constitution of patient changes
4) TSD gene situated on chromosome 15 undergoes mutation

**70.** A woman with two genes for hemophilia and one gene for colour blindness on one of the X' chromosomes marries a normal man, How will the progeny be?
1) 50% haemophilic colour-blind sons and 50% haemophilic sons
2) 50% haemophilic daughters and 50% colour blind daughters
3) All sons and daughters are haemophilic and colour-blind
4) Haemophilic and colour-blind Daughters only.

**71.** In $\beta$ -thalassemia, the affected chromosome is
1) $16^{th}$
2) $14^{th}$
3) $13^{th}$
4) $11^{th}$

**72.** Match the following columns.

| Column I | Column II |
|---|---|

| | | | |
|---|---|---|---|
| A | Sickle-cell anaemia | 1 | Sex - linked |
| B | Colour blindness | 2 | Autosomal chromosome – 7 |
| C | Phenylketonu ria | 3 | Autosomal chromosome -11 |
| D | Cystic fibrosis | 4 | Autosomal chromosome - 12 |

1) A-3, B-4, C-1, D-2
2) A-4, B-1, C-2, D-3
3) A-4, B-1, C-3, D-2
4) A-3, B-1, C-4, D-2

**73.** More men suffer from colour blindness than women because
1) Women are more resistant
2) Male sex hormone testosterone causes the disease
3) The colourblind gene is carried by Ychromosome
4) Men are hemizygous and one defective gene is enough to make colourblind

**74.** Thalassemia is
1) Autosomal recessive disease
2) Autosomal dominant disease
3) Sex-linked dominant disease
4) Sex-linked recessive disease

**75.** I. Haemophilia II. Cystic fibrosis III. Sickle-cell anaemia
IV. Colour blindness V. Cancer VI. Plague
VII. Phenylketonuria VIII. Thalassemia
Choose the correct options for Mendelian disorders:
1) I, II, III, IV, VI, VIII
2) I, II, III, IV, VII, VIII
3) I, II, III, IV, V, VI
4) I, II, III, IV, V, VIII

Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1)** | 3 | **16)** | 1 | **31)** | 2 | **46)** | 2 | **61)** | 1 |
| **2)** | 4 | **17)** | 3 | **32)** | 1 | **47)** | 1 | **62)** | 1 |
| **3)** | 1 | **18)** | 4 | **33)** | 3 | **48)** | 4 | **63)** | 1 |
| **4)** | 2 | **19)** | 1 | **34)** | 4 | **49)** | 4 | **64)** | 4 |
| **5)** | 3 | **20)** | 3 | **35)** | 3 | **50)** | 2 | **65)** | 3 |
| **6)** | 4 | **21)** | 4 | **36)** | 3 | **51)** | 3 | **66)** | 3 |
| **7)** | 1 | **22)** | 3 | **37)** | 3 | **52)** | 2 | **67)** | 3 |
| **8)** | 1 | **23)** | 4 | **38)** | 1 | **53)** | 1 | **68)** | 2 |
| **9)** | 1 | **24)** | 2 | **39)** | 1 | **54)** | 1 | **69)** | 2 |
| **10)** | 2 | **25)** | 1 | **40)** | 2 | **55)** | 3 | **70)** | 1 |
| **11)** | 2 | **26)** | 2 | **41)** | 1 | **56)** | 3 | **71)** | 4 |
| **12)** | 2 | **27)** | 2 | **42)** | 2 | **57)** | 2 | **72)** | 4 |
| **13)** | 4 | **28)** | 2 | **43)** | 2 | **58)** | 2 | **73)** | 4 |
| **14)** | 3 | **29)** | 3 | **44)** | 4 | **59)** | 4 | **74)** | 1 |
| **15)** | 3 | **30)** | 3 | **45)** | 1 | **60)** | 1 | **75)** | 2 |